Li-Ion Cell Charge Manegement

File: tracker_board_cell_charge_magagement.kicad_sch

Power Rails

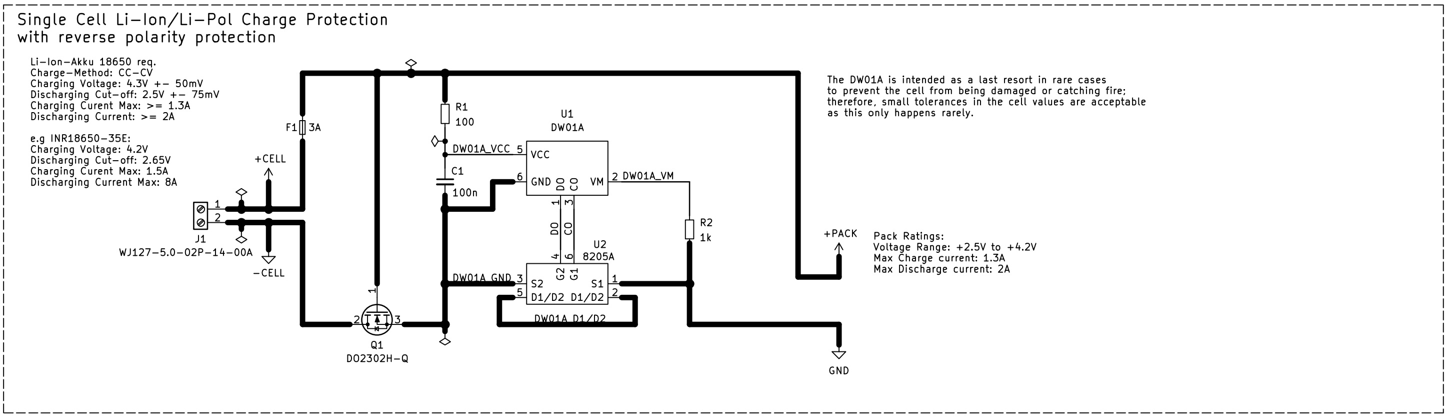
File: tracker_board_power_rails.kicad_sch

Host_MCU

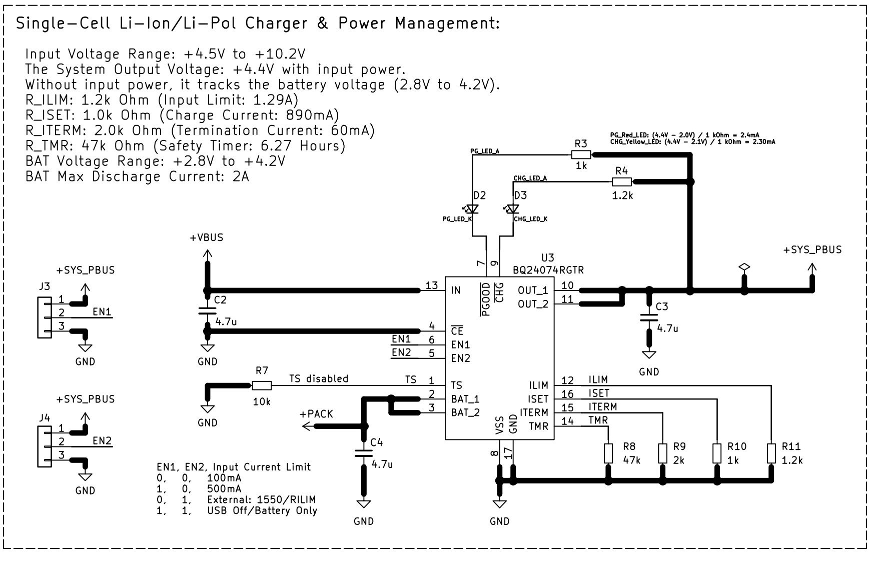
File: host_mcu.kicad_sch

NB_IoT_Modem_p1

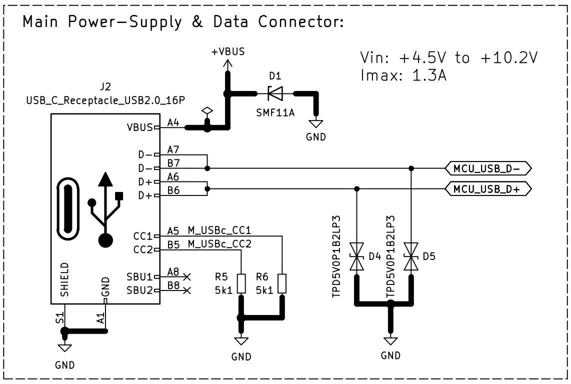
File: lte_modem.kicad_sch



Single Cell Li–Ion/Li–Pol Charge Protection
with reverse polarity protection
Li–Ion–Akku 18650 req.
Charge–Method: CC–CV
Charging Voltage: 4.3V +– 50mV
Discharging Cut–off: 2.5V +– 75mV
Charging Curent Max: >= 1.3A
Discharging Current: >= 2A
e.g INR18650–35E:
Charging Voltage: 4.2V
Discharging Cut–off: 2.65V
Charging Curent Max: 1.5A
Discharging Current Max: 8A
The DW01A is intended as a last resort in rare cases
to prevent the cell from being damaged or catching fire;
therefore, small tolerances in the cell values are acceptable
as this only happens rarely.
Pack Ratings:
Voltage Range: +2.5V to +4.2V
Max Charge current: 1.3A
Max Discharge current: 2A
F1 3A
R1 100
U1 DW01A
DW01A_VCC
VCC
GND
VM
DO
CO
DW01A_VM
C1 100n
R2 1k
U2 8205A
G2
G1
S2
S1
D1/D2
DW01A_GND
DW01A_D1/D2
+CELL
−CELL
J1
WJ127–5.0–02P–14–00A
Q1
DO2302H–Q
+PACK
GND

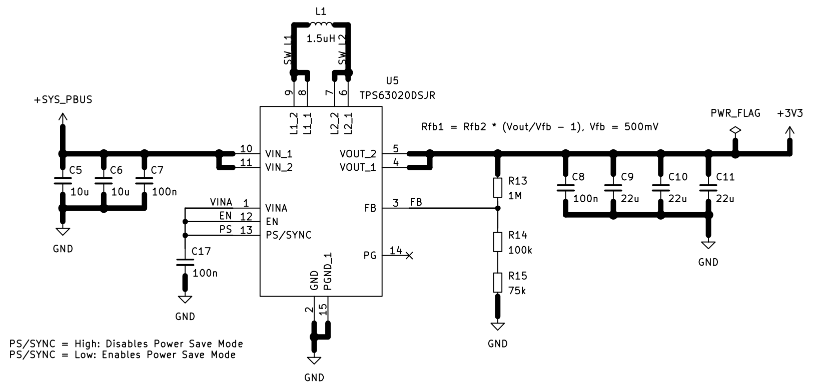


Single–Cell Li–Ion/Li–Pol Charger & Power Management:
Input Voltage Range: +4.5V to +10.2V
The System Output Voltage: +4.4V with input power.
Without input power, it tracks the battery voltage (2.8V to 4.2V).
R_ILIM: 1.2k Ohm (Input Limit: 1.29A)
R_ISET: 1.0k Ohm (Charge Current: 890mA)
R_ITERM: 2.0k Ohm (Termination Current: 60mA)
R_TMR: 47k Ohm (Safety Timer: 6.27 Hours)
BAT Voltage Range: +2.8V to +4.2V
BAT Max Discharge Current: 2A
PG_Red_LED: (4.4V – 2.0V) / 1 kOhm = 2.4mA
CHG_Yellow_LED: (4.4V – 2.1V) / 1 kOhm = 2.30mA
R3 1k
R4 1.2k
PG_LED_A
CHG_LED_A
D2
D3
PG_LED_K
CHG_LED_K
U3
BQ24074RGTR
IN
PGOOD
CHG
OUT_1
OUT_2
CE
EN1
EN2
TS
BAT_1
BAT_2
VSS
GND
ILIM
ISET
ITERM
TMR
+VBUS
+SYS_PBUS
C2 4.7u
C3 4.7u
C4 4.7u
R7 10k
TS disabled
+PACK
R8 47k
R9 2k
R10 1k
R11 1.2k
J3
J4
EN1
EN2
GND
EN1, EN2, Input Current Limit
0, 0, 100mA
1, 0, 500mA
0, 1, External: 1550/RILIM
1, 1, USB Off/Battery Only

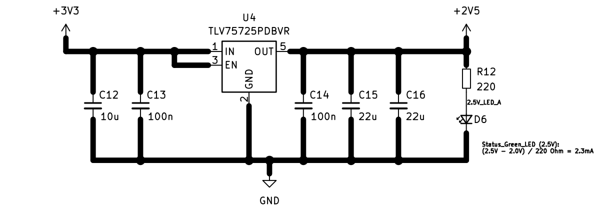


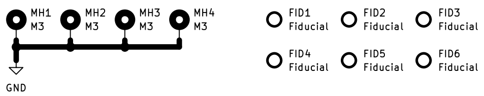
Main Power–Supply & Data Connector:
Vin: +4.5V to +10.2V
Imax: 1.3A
+VBUS
D1
SMF11A
J2
USB_C_Receptacle_USB2.0_16P
VBUS
D−
D+
CC1
CC2
SBU1
SBU2
SHIELD
GND
M_USBc_CC1
M_USBc_CC2
R5 5k1
R6 5k1
MCU_USB_D−
MCU_USB_D+
TPD5V0P1B2LP3
D4
D5
GND

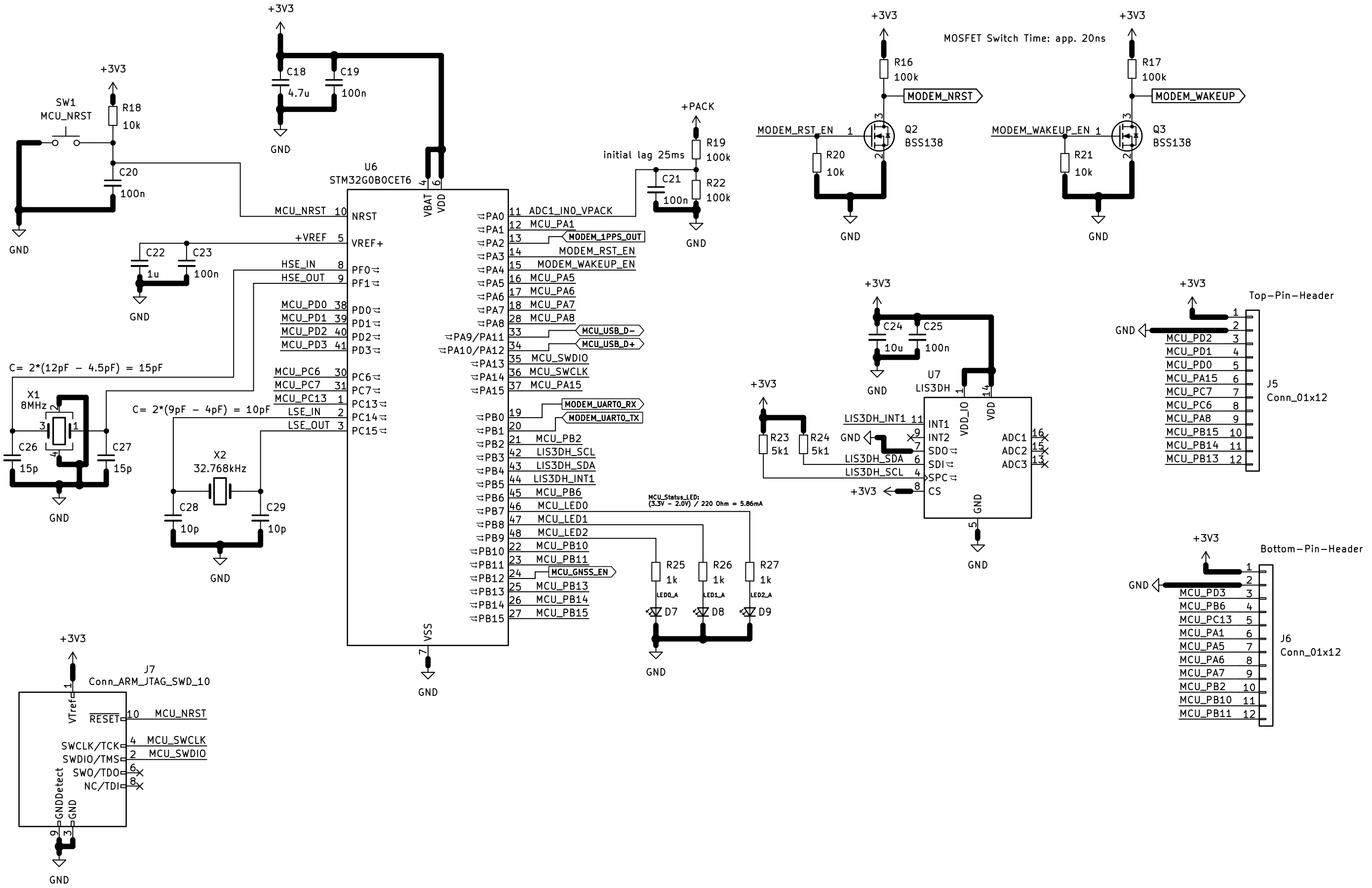
Buck–Boost Converter:
Input voltage range: 2.8V to 4.4V
Output voltage: +3.3V
Iout_max = 2A in Boost–Mode, 4A in Buck–Mode



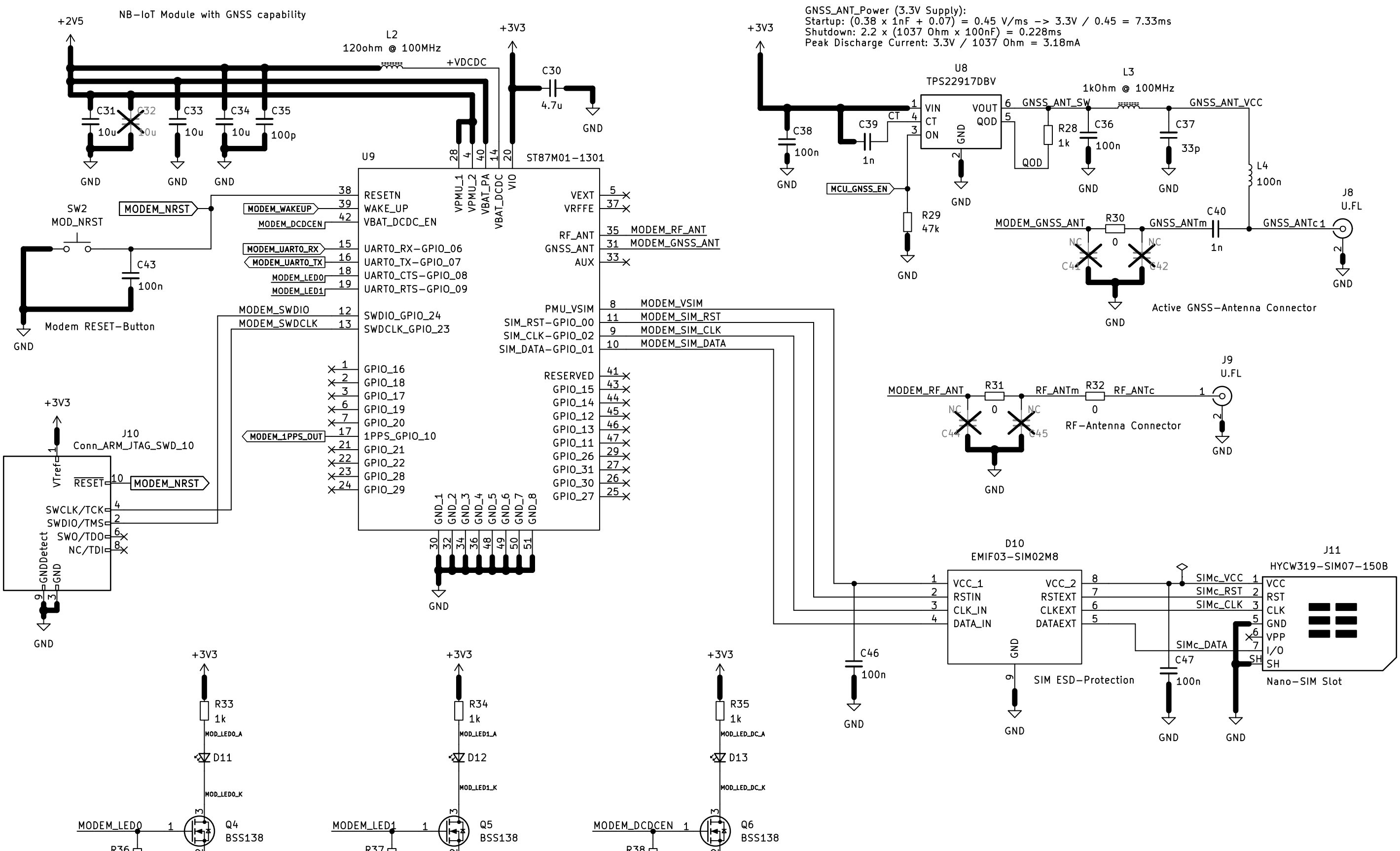
LDO Regulator:
Input voltage: +3.3V
Output voltage: +2.5V
Iout_max = 1A







+3V3
C18
4.7u
C19
100n
GND
SW1
MCU_NRST
R18
10k
C20
100n
U6
STM32G0B0CET6
MCU_NRST
NRST
+VREF
VREF+
C22
1u
C23
100n
HSE_IN
HSE_OUT
PF0
PF1
MCU_PD0
MCU_PD1
MCU_PD2
MCU_PD3
PD0
PD1
PD2
PD3
C= 2*(12pF - 4.5pF) = 15pF
X1
8MHz
C26
15p
C27
15p
MCU_PC6
MCU_PC7
MCU_PC13
LSE_IN
LSE_OUT
PC6
PC7
PC13
PC14
PC15
C= 2*(9pF - 4pF) = 10pF
X2
32.768kHz
C28
10p
C29
10p
VBAT
VDD
VSS
+PACK
R19
100k
R22
100k
C21
100n
initial lag 25ms
ADC1_IN0_VPACK
MCU_PA1
MODEM_1PPS_OUT
MODEM_RST_EN
MODEM_WAKEUP_EN
MCU_PA5
MCU_PA6
MCU_PA7
MCU_PA8
MCU_USB_D-
MCU_USB_D+
MCU_SWDIO
MCU_SWCLK
MCU_PA15
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9/PA11
PA10/PA12
PA13
PA14
PA15
MODEM_UART0_RX
MODEM_UART0_TX
MCU_PB2
LIS3DH_SCL
LIS3DH_SDA
LIS3DH_INT1
MCU_PB6
MCU_LED0
MCU_LED1
MCU_LED2
MCU_PB10
MCU_PB11
MCU_GNSS_EN
MCU_PB13
MCU_PB14
MCU_PB15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
MCU_Status_LED:
(3.3V - 2.0V) / 220 Ohm = 5.86mA
R25
1k
R26
1k
R27
1k
LED0_A
LED1_A
LED2_A
D7
D8
D9
MOSFET Switch Time: app. 20ns
R16
100k
MODEM_NRST
MODEM_RST_EN
Q2
BSS138
R20
10k
R17
100k
MODEM_WAKEUP
MODEM_WAKEUP_EN
Q3
BSS138
R21
10k
C24
10u
C25
100n
U7
LIS3DH
VDD_IO
VDD
R23
5k1
R24
5k1
LIS3DH_INT1
INT1
INT2
SDO
SDI
SPC
CS
LIS3DH_SDA
LIS3DH_SCL
ADC1
ADC2
ADC3
Top-Pin-Header
J5
Conn_01x12
MCU_PD2
MCU_PD1
MCU_PD0
MCU_PA15
MCU_PC7
MCU_PC6
MCU_PA8
MCU_PB15
MCU_PB14
MCU_PB13
Bottom-Pin-Header
J6
Conn_01x12
MCU_PD3
MCU_PB6
MCU_PC13
MCU_PA1
MCU_PA5
MCU_PA6
MCU_PA7
MCU_PB2
MCU_PB10
MCU_PB11
J7
Conn_ARM_JTAG_SWD_10
VTref
RESET
MCU_NRST
SWCLK/TCK
MCU_SWCLK
SWDIO/TMS
MCU_SWDIO
SWO/TDO
NC/TDI
GNDDetect
GND



NB-IoT Module with GNSS capability
Modem RESET-Button
GNSS_ANT_Power (3.3V Supply):
Startup: (0.38 x 1nF + 0.07) = 0.45 V/ms -> 3.3V / 0.45 = 7.33ms
Shutdown: 2.2 x (1037 Ohm x 100nF) = 0.228ms
Peak Discharge Current: 3.3V / 1037 Ohm = 3.18mA
U9 ST87M01-1301
U8 TPS22917DBV
D10 EMIF03-SIM02M8
J11 HYCW319-SIM07-150B
J10 Conn_ARM_JTAG_SWD_10
Active GNSS-Antenna Connector
RF-Antenna Connector
SIM ESD-Protection
Nano-SIM Slot